

Practical – 10

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Roll no: SYB-49

Aim: To understand and implement version control using Git and GitHub.

Theory

Version Control is a system that helps track changes in files over time, allowing multiple users to work on a project while maintaining a complete history of changes.

- **Git:** A distributed version control system used to track changes locally.
- **GitHub:** An online platform used to store and manage Git repositories remotely.
- **Key Functions:** Track file changes, collaborate with others, maintain version history, and restore previous versions.

Step-by-Step Procedure

Step 1: Installation Verification

Verify that Git is installed on your MacBook.

- **Command:** `git --version`

Step 2: Configure Username & Email

Set up your professional identity for your commits.

- **Commands:** * `git config --global user.name "Sanyogita"`
 - `git config --global user.email "somwanshisanyogita@gmail.com"`

Step 3: Create Local Repository

Initialize a new project folder on your MacBook.

- **Commands:** * `mkdir PDS`
 - `cd PDS`
 - `git init`

Step 4: Create a new text file to track.

- **Command:** `echo "Hello Git" > file1.txt`

Step 5: Add File to Staging Area.

Prepare the file for a commit.

- **Command:** `git add file1.txt`

Step 6: Commit Changes

Save your changes permanently in the local repository.

- **Command:** `git commit -m "Initial commit"`

Step 7: Link and Push to GitHub

Connect your local "PDS" folder to your GitHub repository ("PDSPRACTICAL1to9") and upload the files.

- **Commands:** * `git remote add origin`
<https://github.com/gita1804/PDSPRACTICAL1to9.git>
 - `git branch -M main`
 - `git push -u origin main`

```
PDS — -zsh — 102x56

Last login: Thu May 7 13:07:41 on ttys000
[sanyogita@sanyogitas-MacBook-Air ~ % cd ~/PDS
[sanyogita@sanyogitas-MacBook-Air PDS % rm -rf .git
[sanyogita@sanyogitas-MacBook-Air PDS % git init
Initialized empty Git repository in /Users/sanyogita/PDS/.git/
[sanyogita@sanyogitas-MacBook-Air PDS % git config --global user.name "Sanyogita"
[sanyogita@sanyogitas-MacBook-Air PDS % git config --global user.email "somwanshisanyogita@gmail.com"
[sanyogita@sanyogitas-MacBook-Air PDS % echo "Hello Git" > file1.txt
[sanyogita@sanyogitas-MacBook-Air PDS % git add file1.txt
[sanyogita@sanyogitas-MacBook-Air PDS % git commit -m "Initial commit"
[main (root-commit) b4d04b7] Initial commit
 1 file changed, 1 insertion(+)
 create mode 100644 file1.txt
[sanyogita@sanyogitas-MacBook-Air PDS % git log
commit b4d04b7900f67144db0321f886304718d7bc216a (HEAD -> main)
Author: Sanyogita <somwanshisanyogita@gmail.com>
Date: Thu May 7 14:04:54 2026 +0530

    Initial commit
[sanyogita@sanyogitas-MacBook-Air PDS % git remote add origin https://github.com/gita1804/PDSPractical1to9.git
[sanyogita@sanyogitas-MacBook-Air PDS % git branch -M main
[sanyogita@sanyogitas-MacBook-Air PDS % git push -u origin main
Username for 'https://github.com': gita1804
Password for 'https://gita1804@github.com':
To https://github.com/gita1804/PDSPractical1to9.git

[sanyogita@sanyogitas-MacBook-Air PDS % git pull origin main --rebase
remote: Enumerating objects: 14, done.
remote: Counting objects: 100% (14/14), done.
remote: Compressing objects: 100% (13/13), done.
remote: Total 14 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (14/14), 9.59 MiB | 771.00 KiB/s, done.
From https://github.com/gita1804/PDSPractical1to9
 * branch          main      -> FETCH_HEAD
 * [new branch]    main      -> origin/main
Successfully rebased and updated refs/heads/main.
[sanyogita@sanyogitas-MacBook-Air PDS % git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 292 bytes | 292.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/gita1804/PDSPractical1to9.git
    db1d127..4a6b0e8  main -> main
sanyogita@sanyogitas-MacBook-Air PDS %
```

Questions & Answers

1. What is the difference between Git and GitHub?

- Git:** A version control system used to track changes in files locally.
- GitHub:** A cloud-based platform used to store and manage Git repositories online.

2. Difference between git add and git commit?

- git add:** Adds changes to the staging area.
- git commit:** Saves changes permanently in the repository.

3. What is the staging area in Git?

- a. The staging area is a temporary space where changes are prepared before committing them to the repository.

4. Difference between git push and git pull?

- a. **git push:** Uploads local changes to a GitHub repository.
- b. **git pull:** Downloads the latest changes from GitHub to the local system.

Conclusion

This practical successfully demonstrated the use of Git and GitHub for version control on macOS. All essential steps—including repository initialization, staging, committing, and pushing to a remote repository—were performed to manage project history and enable collaboration.

The screenshot shows a GitHub repository page for 'PDSPractical1to9' by user 'gita1804'. The repository is public and has 2 commits. The file list shows several PDF files (EX1_B3_49_Sanyogita_Somwanshi.pdf to EX9_B3_49_Sanyogita_Somwanshi.pdf) and a README.md file, all added 20 minutes ago except for the initial commit of README.md which was 24 minutes ago. The README content is visible at the bottom, titled 'PDSPractical1to9' and describing it as 'Practical submissions for PDS, including PDF reports and documentation for Exercise 1 to 9.' The right sidebar shows repository statistics: 0 stars, 0 watching, 0 forks, and 1 contributor (gita1804 sanyogita_somwanshi).

File Name	Commit Message	Time Ago
EX1_B3_49_Sanyogita_Somwanshi.pdf	Add PDS Practicals 1-9	20 minutes ago
EX2_B3_49_Sanyogita_Somwanshi.pdf	Add PDS Practicals 1-9	20 minutes ago
EX3_B3_49_Sanyogita_Somwanshi.pdf	Add PDS Practicals 1-9	20 minutes ago
EX4_B3_49_Sanyogita_Somwanshi.pdf	Add PDS Practicals 1-9	20 minutes ago
EX5_B3_49_Sanyogita_Somwanshi.pdf	Add PDS Practicals 1-9	20 minutes ago
EX6_B3_49_Sanyogita_Somwanshi.pdf	Add PDS Practicals 1-9	20 minutes ago
EX7_B3_49_Sanyogita_Somwanshi.pdf	Add PDS Practicals 1-9	20 minutes ago
EX8_B3_49_Sanyogita_Somwanshi.pdf	Add PDS Practicals 1-9	20 minutes ago
EX9_B3_49_Sanyogita_Somwanshi.pdf	Add PDS Practicals 1-9	20 minutes ago
README.md	Initial commit	24 minutes ago



sanyogita_somwanshi
gita1804 · she/her

Edit profile

4 followers · 5 following

Find a repository...TypeSortNew

- PDSPractical1to9Public

Practical submissions for PDS, including PDF reports and documentation for Exercise 1 to 9.

Updated 10 minutes ago

Star
- 2.0Public

Updated last week

Star
- SkillSyncPublic

Config files for my GitHub profile.

[config](#) [github-config](#)

Updated on Mar 4, 2022

Star